AI Chatbot & Routine Management for Postpartum Depression Support

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Project Proposal Report

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DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

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Signature of the supervisor: Date:

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Co - Signature of the supervisor: Date:

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I would like to take this opportunity to express my sincere gratitude to everyone who has supported the development of this research project proposal on the AI-Based Perinatal Depression Screening and Support System.

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ABSTRACT

Postpartum Depression (PPD) is a significant maternal mental health condition affecting mothers during pregnancy and after childbirth. Globally, approximately 10–20% of mothers experience postpartum mental health disorders. In Sri Lanka, studies report that the prevalence of PPD is about 15.5% within the first 10 days after childbirth and 7.8% after four weeks, indicating that many mothers experience psychological distress during the early postpartum period. Despite the seriousness of this issue, many women do not receive adequate support due to mental health stigma, financial constraints, limited access to professional care, and insufficient availability of mental health resources. Recent studies have explored the use of Artificial Intelligence–based chatbots for mental health support; however, several research gaps remain. Conversational AI solutions for perinatal mental health are largely unexplored in low- and middle-income countries such as Sri Lanka, and most existing systems are designed for Western healthcare environments. Additionally, many digital mental health tools rely on subscription models, insurance coverage, or specialized clinical infrastructure, making them less accessible to mothers in developing regions. Furthermore, most existing PPD chatbots focus only on conversational support and do not provide structured daily guidance or personalized routines that help mothers manage symptoms in daily life. Many systems also provide uniform responses without adapting to the user's risk level of postpartum depression. To address these limitations, this research proposes a mobile-based AI support system that integrates a conversational chatbot with postpartum depression screening and personalized routine generation. The chatbot will use a Retrieval-Augmented Generation (RAG) architecture to provide reliable responses. The system will be developed using React Native to ensure accessibility. The expected outcome is a fully functional application that encourages mothers to follow supportive routines and improve postpartum mental well-being, while also enabling hospitals to use the system for patient engagement and follow-up care.

Keywords: Postpartum Depression, AI Chatbot, Retrieval-Augmented Generation, Maternal Mental Health, Mobile Health Application

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LIST OF ABBREVIATIONS

Table 01 – LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
CBT	Cognitive Behavioral Therapy
DSR	Design Science Research
LLM	Large Language Model
LMIC	Low- and Middle-Income Countries
NLP	Natural Language Processing
PPD	Postpartum Depression
RAG	Retrieval-Augmented Generation
SDG	Sustainable Development Goal
WHO	World Health Organization

1. INTRODUCTION

1.1 Background & Context

Postpartum Depression (PPD) is a serious maternal mental health condition that affects women during pregnancy and after childbirth. It is characterized by symptoms such as persistent sadness, anxiety, fatigue, sleep disturbances, and difficulty bonding with the newborn. The early weeks after childbirth are particularly challenging because mothers experience hormonal changes, physical recovery from delivery, sleep deprivation, and the responsibility of caring for an infant. Without proper support, these challenges can negatively affect both the emotional well-being of mothers and the healthy development of their babies.

This issue mainly affects pregnant women and mothers during the early postpartum period, especially within the first four weeks after childbirth. During this stage, mothers often experience emotional stress while adapting to new responsibilities and lifestyle changes. Many mothers also spend significant time on social media platforms such as Facebook and Instagram during pregnancy and postpartum periods. However, these platforms mainly provide entertainment rather than meaningful mental health support, highlighting the need for a dedicated digital platform designed for maternal well-being.

Maternal mental health disorders are recognized as a major global public health concern. According to the World Health Organization, approximately 10% of pregnant women and 13% of postpartum women worldwide experience mental health disorders, including postpartum depression. In Sri Lanka, studies indicate that the prevalence of postpartum depression is about 15.5% within the first 10 days after childbirth and around 7.8% after four weeks postpartum, showing that many mothers experience psychological distress during the early postpartum period.

Several digital mental health tools and chatbot-based systems such as Woebot, Wysa, and Moment for Parents have been developed to provide emotional support through conversational interactions. However, many of these systems are designed primarily for users in developed countries and are not adapted to the cultural and social context of countries such as Sri Lanka. Some applications also require paid subscriptions or access through healthcare insurance, making them inaccessible for many users.

To address these limitations, this research proposes an AI-powered mobile-based support system designed for pregnant women and mothers in the early postpartum period. The system will include a conversational chatbot that provides emotional support, reliable maternal health information, and personalized guidance. The chatbot will use a Retrieval-Augmented Generation (RAG) approach to generate responses from verified knowledge sources while reducing misinformation risks in healthcare contexts.

This research aligns with United Nations Sustainable Development Goal 3 (Good Health and Well-being) by supporting maternal mental health. It also contributes to the Artificial Intelligence and Digital Health research cluster, integrating conversational AI, natural language processing, and knowledge retrieval techniques to develop an intelligent healthcare support system.

1.2 Literature Review

The growing prevalence of postpartum depression (PPD) has made maternal mental health an important public health concern worldwide. Postpartum depression is a mood disorder that affects women during pregnancy or within the first year after childbirth and can lead to symptoms such as sadness, anxiety, fatigue, and sleep disturbances. These symptoms can negatively affect both the mother and the newborn, particularly during the early postpartum period. Studies indicate that postpartum depression affects a significant proportion of mothers globally and requires early detection and continuous emotional support to reduce its impact on maternal and infant health outcomes [1].

Several research studies have explored the use of artificial intelligence and digital health technologies to support mental health care. AI-powered conversational agents and chatbots are increasingly used to provide emotional support, mental health guidance, and cognitive behavioral therapy (CBT)-based interventions through natural language conversations. A scoping review of mental health chatbots identified more than forty chatbot systems designed for psychological support, including well-known applications such as Woebot and Wysa. These systems demonstrate the potential of conversational AI to provide scalable and accessible mental health assistance for individuals experiencing depression and anxiety [2],

[3].

In the context of postpartum mental health, researchers have begun exploring specialized chatbot systems to support mothers during pregnancy and the postpartum period. For example, the Woebot chatbot delivers CBT-based conversational therapy to women experiencing postpartum depression and has shown improvements in depressive symptoms in clinical studies [4]. Similarly, conversational AI platforms such as Wysa have been used to provide emotional support and self-help strategies for users experiencing maternal mental health challenges through mobile applications [5]. These systems demonstrate that conversational agents can play an important role in providing continuous emotional support and mental health monitoring [6].

Despite these advancements, several limitations remain in existing digital mental health solutions. Many chatbot systems are designed primarily for general mental health conditions and are not specifically tailored to the unique needs of postpartum mothers. In addition, many applications are developed in high-income countries and are not adapted to the cultural, social, or linguistic context of low- and middle-income countries. Access to mental health support also remains limited in many regions due to financial barriers, stigma, and shortages of mental health professionals [7], [8].

Furthermore, several studies highlight the need for reliable and context-aware AI systems that can provide evidence-based responses while minimizing the risk of misinformation in sensitive healthcare contexts. The integration of knowledge-grounded conversational systems, such as retrieval-augmented generation (RAG) models, has been suggested as a promising approach to improve the reliability and safety of AI-generated responses in mental health support systems. By combining conversational AI with verified knowledge sources, these systems can provide more accurate, personalized, and context-aware support for postpartum mothers [9], [10].

Recent studies have also explored the effectiveness of conversational agents in delivering cognitive behavioral therapy (CBT) interventions and emotional support for individuals experiencing depression and anxiety [14], [18].

1.3 Research Gap

Although several digital mental health tools and AI chatbots have been developed to provide psychological support, there is a lack of AI-driven systems specifically designed for postpartum depression support in low- and middle-income countries such as Sri Lanka. Most existing chatbot systems focus on general mental health conditions and do not address the unique emotional and practical challenges faced by pregnant women and mothers during the early postpartum period.

Current chatbot-based solutions such as Woebot, Wysa, and Moment for Parents provide conversational support but have several limitations. Many of these systems are designed for users in high-income countries and require subscription services or healthcare insurance access. Additionally, most existing systems provide general responses and are not designed to adapt their behavior based on postpartum depression risk levels or culturally relevant maternal healthcare needs.

Another limitation in current research is the lack of integration between postpartum depression screening systems and conversational AI support tools. Most chatbot solutions operate independently and do not adjust their responses based on the severity of a user's mental health condition. This reduces their effectiveness in providing personalized and context-aware support for mothers experiencing different levels of psychological distress.

Therefore, this research addresses these gaps by proposing an AI-powered mobile-based support system that integrates postpartum depression risk assessment with a conversational chatbot. The system uses a Retrieval-Augmented Generation (RAG) approach to generate reliable responses from verified knowledge sources while providing personalized guidance and emotional support for pregnant and postpartum mothers. This approach aims to improve accessibility, reliability, and contextual relevance of digital maternal mental health support systems.

Table 02 – System Comparison –Existing chatbots and systems.

Features	Woebot	Wysa	MomCare	Generic Chatbot	Our System
Focus on Postpartum Depression	✗	✗	✓	✗	✓
AI Conversational Emotional Support	✓	✓	✓	✓	✓
Evidence-based knowledge retrieval (RAG)	✗	✗	✗	✓	✓
Personalized daily routines	✗	✗	✗	✗	✓
Adaptive responses based on PPD risk level	✗	✗	✗	✗	✓
Designed for developing countries	✗	✗	✗	✓	✓
Maternal mental health knowledge base	✗	✗	✓	✓	✓

2 OBJECTIVES

2.1 Main Objective

To design and develop an AI-powered conversational chatbot integrated into a mobile application that supports pregnant women and mothers in the early postpartum period by providing reliable maternal mental health guidance. The system will utilize a Retrieval-Augmented Generation (RAG) approach to generate context-aware and knowledge-grounded responses related to postpartum depression. By integrating postpartum depression risk-level information from the screening module, the chatbot will deliver personalized emotional support, educational guidance, and structured daily recommendations aimed at improving maternal well-being. The proposed solution seeks to enhance accessibility to mental health assistance, encourage positive daily routines, and provide a supportive digital environment that mothers can use as a healthier alternative to general social media platforms during pregnancy and the postpartum period.

2.2 Specific Objectives

To develop a domain-specific knowledge base for postpartum mental health support: Collect, organize, and structure reliable information related to postpartum depression, maternal mental health care, coping strategies, and self-care guidelines from scholarly articles, medical resources, and verified healthcare guidelines. Prepare and preprocess this knowledge so it can be efficiently retrieved and used by the chatbot system. The structured knowledge base will ensure that the AI assistant provides accurate, relevant, and evidence-based responses when interacting with pregnant women and postpartum mothers.

To design and implement a Retrieval-Augmented Generation (RAG) based chatbot system:

Develop a conversational AI system that integrates a retrieval mechanism with a language model to generate context-aware responses for maternal mental health queries. The RAG framework will retrieve relevant information from the knowledge base before generating responses, ensuring that the chatbot produces reliable and fact-based answers. This approach helps reduce hallucination issues commonly found in generative AI systems and improves the safety and reliability of the chatbot when addressing sensitive postpartum mental health topics.

To develop adaptive conversational responses based on postpartum depression risk levels:

Design a chatbot interaction mechanism that adjusts its responses according to the user's postpartum depression risk level obtained from the screening module. The chatbot should provide informational and motivational responses for low-risk users, structured guidance and stronger encouragement for medium-risk users, and supportive crisis-sensitive guidance for high-risk users. This adaptive behavior will allow the chatbot to provide personalized support and ensure that users receive appropriate guidance depending on their emotional and psychological condition.

To develop a mobile-based conversational support application:

Implement the chatbot system within a mobile application environment using modern mobile development frameworks such as React Native. The application will provide an accessible and user-friendly interface that allows pregnant women and postpartum mothers to easily interact with the AI assistant. The system should support continuous interaction, provide guidance for daily well-being, and act as a supportive digital platform that encourages healthier engagement compared to general social media usage.

To ensure ethical, privacy, and safety considerations in maternal mental health support:

Design the chatbot system with strong ethical and privacy considerations when handling sensitive maternal mental health information. Implement mechanisms to protect user data privacy, maintain confidentiality, and ensure responsible use of collected information. The system should also include safeguards such as crisis-sensitive responses and recommendations for professional medical help when severe postpartum depression symptoms are detected.

3 METHODOLOGY

3.1 Research Approach

This research follows a Design Science Research (DSR) approach, which focuses on designing and developing an innovative technological solution to address a real-world problem. The study aims to design and implement an AI-powered conversational chatbot that supports pregnant women and mothers experiencing postpartum depression. The research involves developing the system, integrating relevant technologies, and evaluating its effectiveness through experimental testing. The iterative design process will allow improvements based on testing results and feedback to ensure the system provides reliable and context-aware maternal mental health support.

3.2 Technical Methods and Algorithms

The proposed system will utilize a Retrieval-Augmented Generation (RAG) architecture to generate accurate and context-aware responses. The RAG framework combines a retrieval mechanism with a generative language model. When a user submits a query, the system first converts the input into vector embeddings using a sentence embedding model. A similarity search algorithm retrieves the most relevant information from a curated knowledge base related to postpartum depression. The retrieved information is then provided to a language model to generate a reliable response. This method helps reduce hallucinations and ensures that the chatbot responses are grounded in verified medical knowledge. Additionally, a response adaptation mechanism will be implemented to adjust chatbot responses according to postpartum depression risk levels obtained from the screening module.

3.3 Tools and Platforms with Justification

The proposed system will be implemented using modern AI and mobile development technologies. The chatbot interface will be developed as a mobile application using React Native, allowing the system to run on both Android and iOS platforms while maintaining a consistent user experience. The backend services will be developed using Python, which provides strong support for machine learning and natural language processing libraries.

Vector embeddings and similarity search will be implemented using vector database tool Pinecone, which enable efficient retrieval of relevant information for the RAG architecture. These technologies are chosen due to their scalability, reliability, and strong integration capabilities with AI-based systems.

3.4 Data Collection and Ethical Considerations

The knowledge base for the chatbot will be developed using information collected from scholarly articles, maternal health guidelines, and postpartum depression research studies. Additional anonymized user interaction data may be collected to improve the system's conversational responses and performance. Since the system deals with sensitive maternal mental health information, strict ethical considerations will be followed. Personal user data will not be publicly shared, and user privacy will be protected through secure data storage and anonymization techniques. The system will also include clear disclaimers indicating that the chatbot is a supportive tool and does not replace professional medical advice.

3.5 Validation Strategy and Performance Metrics

The effectiveness of the proposed chatbot system will be evaluated through both technical and qualitative performance measures. Retrieval performance will be measured using metrics such as precision and recall to evaluate the relevance of retrieved knowledge base responses. Response quality will be assessed using semantic similarity metrics and human evaluation to determine the accuracy, relevance, empathy, and helpfulness of chatbot responses. Additionally, usability testing will be conducted to evaluate user satisfaction and system usability when interacting with the chatbot within the mobile application environment.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

Overview of the Proposed Solution and Its Components

The proposed system is an AI-powered mobile assistant designed to support pregnant women and postpartum mothers experiencing emotional distress related to Postpartum Depression (PPD). It integrates a Retrieval-Augmented Generation (RAG) architecture with a conversational chatbot to provide reliable, context-aware, and empathetic responses based on verified maternal health information.

The system includes several key components: a mobile application interface, chatbot interaction layer, knowledge base and data layer, retrieval pipeline, and response generation module. Users interact with the chatbot through text or voice queries. The system processes the query, retrieves relevant information, validates it through guardrail mechanisms, and generates an appropriate response using a language model. The response is delivered along with supportive features such as personalized routines and reminders.

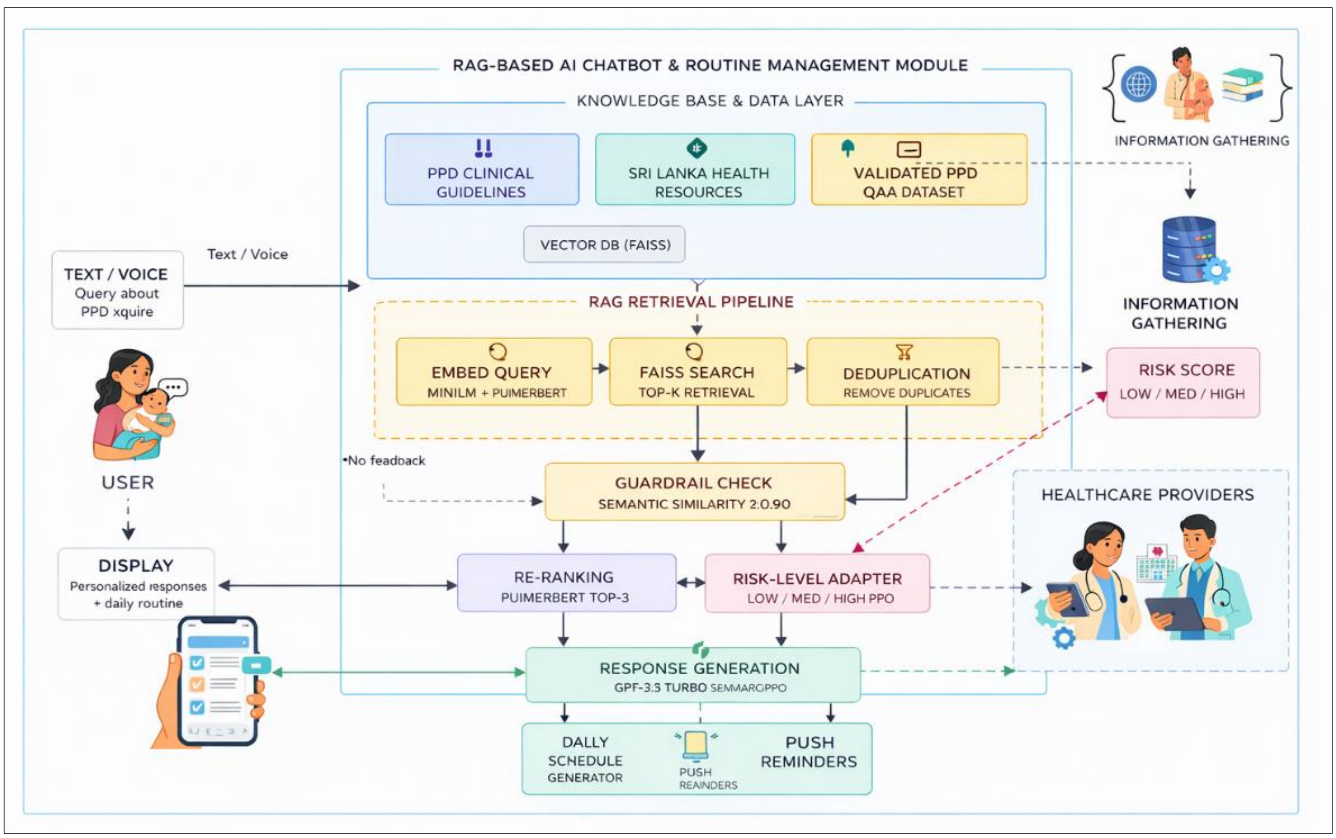

Figure 1 – High-Level System Architecture of the PPD AI Chatbot System

System Diagram of Components

The system architecture is composed of five primary layers: the user interaction layer, the knowledge base and data layer, the retrieval pipeline, the validation and ranking layer, and the response generation layer.

The user interaction layer allows mothers to communicate with the system through the mobile application using natural language queries. The knowledge base layer contains validated postpartum depression information collected from clinical guidelines, Sri Lankan health resources, and curated datasets. These documents are converted into vector embeddings and stored in a vector database using FAISS indexing.

The retrieval pipeline processes user queries by generating embeddings and retrieving semantically similar information from the knowledge base. A guardrail mechanism evaluates whether the query is relevant to postpartum depression and prevents the system from responding to out-of-domain queries. The retrieved responses are then re-ranked using domain-specific embeddings before being passed to the language model for response generation.

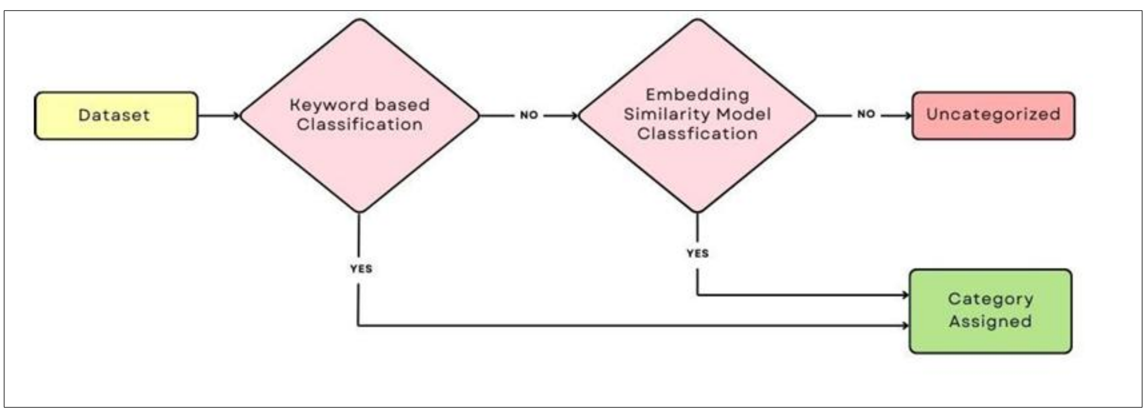

Figure 2 – Hybrid Query Classification Process for Dataset Categorization

Individual Component and Its Parts

The Chatbot Interaction Module forms the core component of the system. It handles user communication and coordinates the retrieval and response generation processes. This module contains several internal components including the query embedding process, semantic search mechanism, response validation system, and response generation module.

The Knowledge Base Component stores domain-specific postpartum depression resources and datasets. These datasets include clinical guidelines, Sri Lankan maternal health resources, and validated PPD question-answer datasets. All textual information is transformed into numerical embeddings using models such as MiniLM and PubMedBERT to enable semantic retrieval.

The Retrieval and Validation Module ensures that the chatbot generates reliable and context-appropriate responses. This module performs semantic similarity checks, duplicate removal, and query classification to determine whether a user query falls within the supported domain.

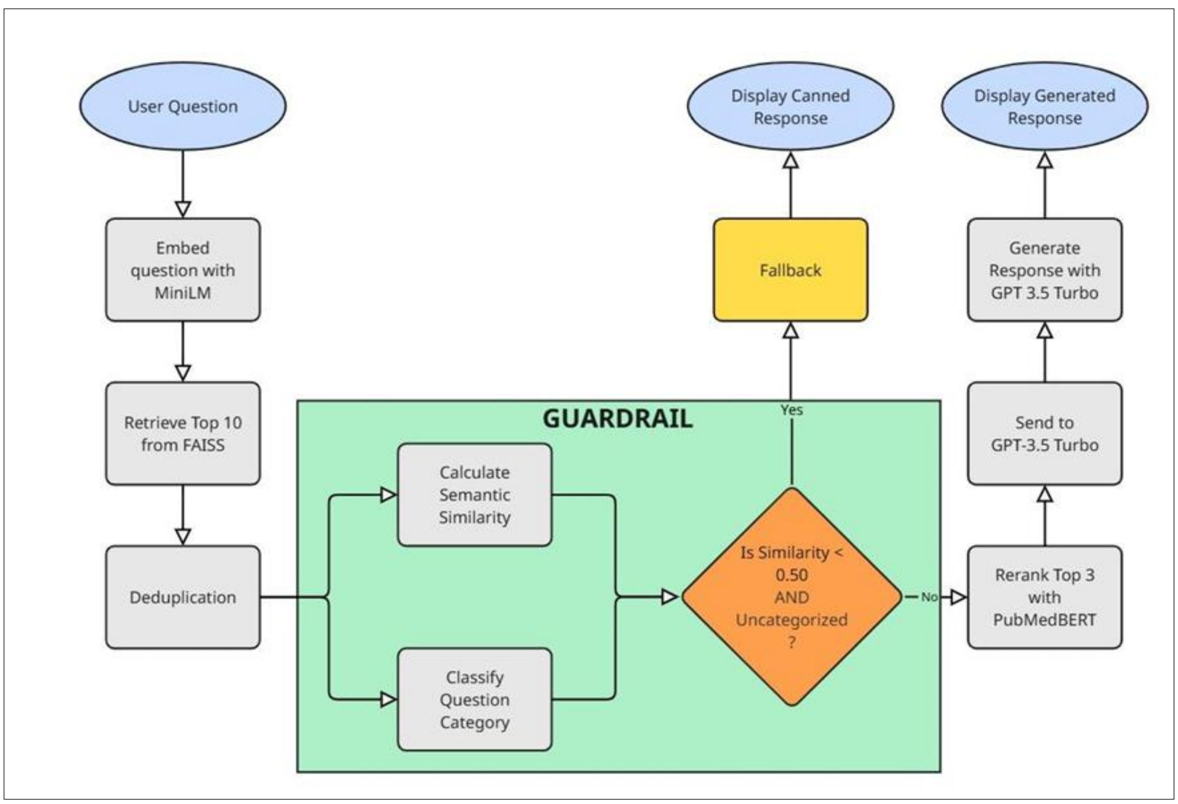

Figure 3 – RAG Response Generation and Guardrail Validation Pipeline

Component Interactions

When a user submits a query through the mobile application, the system converts the text into vector embeddings using a sentence embedding model. These embeddings are used to perform a similarity search in the FAISS vector database to retrieve relevant knowledge entries. The retrieved results pass through guardrail validation to check domain relevance and semantic similarity.

If valid, the system reranks the results using a domain-specific embedding model to identify the most relevant information. This context is then provided to the language model, which generates a response tailored to the user's query. The response is delivered through the mobile interface and may trigger additional features such as daily routine suggestions or push notifications.

Data Flow Explanation

The data flow begins with user input in the mobile application. The query is converted into vector representations and used to retrieve relevant information from the knowledge base. After validation through guardrail checks, the most relevant results are reranked and passed to the language model. The generated response is then returned to the user interface.

Scalability and Security Considerations

The architecture supports scalability through vector databases and cloud-based services that handle large query volumes efficiently. Security is ensured through anonymized data storage, secure communication protocols, and guardrail mechanisms that prevent harmful responses and encourage professional medical consultation when necessary.

5. USER REQUIREMENTS

5.1 Stakeholder identification

The proposed system involves several stakeholders who interact with or benefit from the postpartum mental health support platform.

Primary Stakeholders

- Pregnant women and mothers in the early postpartum period who require emotional support, information, and guidance regarding postpartum mental health.

Secondary Stakeholders

- Healthcare professionals such as doctors, midwives, and maternal health workers who may recommend or use the system to support mothers.
- Researchers and developers involved in improving AI-based maternal mental health technologies.

Tertiary Stakeholders

- Hospitals and maternal healthcare institutions that may adopt the system to assist patients with postpartum mental health monitoring and follow-up support.

5.2 Functional Requirements

User Query Processing

- The system should allow users to enter questions or concerns related to pregnancy, postpartum depression, and maternal well-being.

Conversational Chatbot Interaction

- The system should provide an interactive chatbot interface that enables mothers to communicate with the AI assistant through natural language conversations.

Knowledge Retrieval and Response Generation

- The chatbot should retrieve relevant information from a verified knowledge base and generate reliable responses using the Retrieval-Augmented Generation (RAG) approach.

Risk-Level Based Response Adaptation

- The chatbot should adjust its responses based on the postpartum depression risk level obtained from the screening module.

Personalized Daily Guidance

- The system should provide supportive recommendations and simple daily guidance to improve maternal well-being.

User Feedback Mechanism

- The application should allow users to provide feedback or ratings on chatbot responses to improve system performance.

Mobile Application Interaction

- The system should enable users to access the chatbot through a mobile application interface.

API Integration

- The system should support integration with backend services, knowledge bases, and AI models through APIs.

5.3 Non-Functional Requirements

Performance
The system should generate responses quickly to ensure smooth conversational interaction with minimal delay.

Accuracy and Reliability
The chatbot should provide reliable and evidence-based responses derived from verified maternal health knowledge sources.

Usability and User Experience
The mobile application interface should be simple, intuitive, and easy to use for pregnant women and postpartum mothers.

Security and Privacy
The system must protect sensitive user information and ensure confidentiality of maternal mental health data.

Availability and Reliability
The chatbot system should be available for users at any time to provide continuous support.

Compatibility
The mobile application should be compatible with common mobile platforms such as Android and iOS.

6. COMMERCIALIZATION PLAN

Target Market and Customer Persona

The primary target market for the proposed system includes pregnant women and mothers in the early postpartum period who require accessible mental health guidance and emotional support. These users typically seek convenient, private, and affordable digital solutions that can help them manage postpartum stress and emotional challenges. Secondary customers include hospitals, maternity clinics, midwives, and maternal healthcare organizations that may adopt the system as a supportive tool for monitoring and assisting postpartum mothers. The typical customer persona is a smartphone-using mother aged between 20–40 years who is seeking reliable health guidance while managing daily responsibilities.

Value Proposition

The proposed mobile-based AI assistant provides accessible and personalized postpartum mental health support. Unlike general social media platforms that provide entertainment but no meaningful guidance, this system offers evidence-based information, emotional support, and personalized recommendations using AI technology. The chatbot's use of a Retrieval-Augmented Generation (RAG) approach ensures reliable responses grounded in verified maternal health knowledge sources.

Revenue Model

The system can adopt a hybrid revenue model. Basic chatbot support and maternal mental health guidance can remain free to ensure accessibility for mothers. Additional services such as advanced monitoring features, premium insights for healthcare providers, and institutional dashboards for hospitals can be offered through subscription-based services or institutional partnerships.

Cost Estimation

The main cost components include system development, AI infrastructure, cloud hosting services, maintenance, and continuous updates of the knowledge base. Additional costs may arise from user interface improvements, security enhancements, and system scalability to support a growing number of users.

Pricing Strategy

A freemium pricing strategy can be adopted. The core chatbot functionality will remain free to maximize accessibility for mothers, while premium features designed for healthcare institutions and clinics may be offered through subscription packages or service agreements.

Competitive Advantage

The proposed system offers several competitive advantages compared to existing digital mental health tools. It is designed specifically for postpartum mothers, integrates postpartum depression risk-level adaptation, and utilizes knowledge-grounded AI responses to reduce misinformation.

Intellectual Property Considerations

The system's architecture, chatbot framework, and knowledge integration mechanisms may be protected through intellectual property rights such as software copyrights and potential patents for unique system components. Proper licensing and ethical data usage policies will also be implemented to ensure compliance with data protection regulations and responsible AI development.

Table 03 – Estimated Annual Financial Outcome of the Proposed System

Category	Description	Estimated Value
Total Users (Free)	Mothers using the free chatbot service	5,000 users
Premium Institutional Users	Hospitals / maternity clinics subscribing	10 institutions
Institution Subscription Fee	Monthly subscription per institution	LKR 12,000
Annual Institutional Revenue	10 × 12,000 × 12 months	LKR 1,440,000
Cloud Hosting & AI Infrastructure	Server hosting, AI API usage	- LKR 90,000
System Maintenance	Technical maintenance and updates	- LKR 60,000
Knowledge Base Updates	Updating medical knowledge resources	- LKR 25,000
Security & Data Protection	System security and backups	- LKR 15,000
Total Annual Cost	Infrastructure + maintenance + updates	- LKR 190,000
Estimated Annual Profit	Revenue – Total Cost	LKR 1,250,000

7. BUDGET AND JUSTIFICATION

Hardware Costs

No additional hardware costs are required. The development will be carried out using existing laptops owned by the team members. Personal smartphones will be used for testing the mobile application. Backup storage may be used for storing training data and development files.

Software Licenses

Possible software-related costs include purchasing a domain name for deployment and database hosting services. Additional costs may arise from third-party APIs or AI services used for the chatbot system.

Cloud Services

Cloud services may be required for hosting the backend system, storing the knowledge base, and running the chatbot services. These services will support system availability and scalability.

Human Effort Cost

The project will be developed by the research team as part of the academic research work. Therefore, no direct financial cost is allocated for human effort.

Justification for Each Expense

The budget mainly supports system deployment and service availability. Domain services, cloud infrastructure, and APIs ensure reliable hosting, secure data storage, and proper functioning of the chatbot system.

Table 04 – Budget table

Description	Cost (LKR)
Electricity	5000.00
Internet	5000.00
Communication	2000.00
Transportation (Hospital Visits & Data Collection)	4000.00
Maternal Health Consultation / Expert Review	6000.00
Paper Publishing	15000.00
Backend Hosting	12000.00
Domain & Software Services	6500.00
Total	55500.00

8. WORK BREAKDOWN STRUCTURE (WBS)

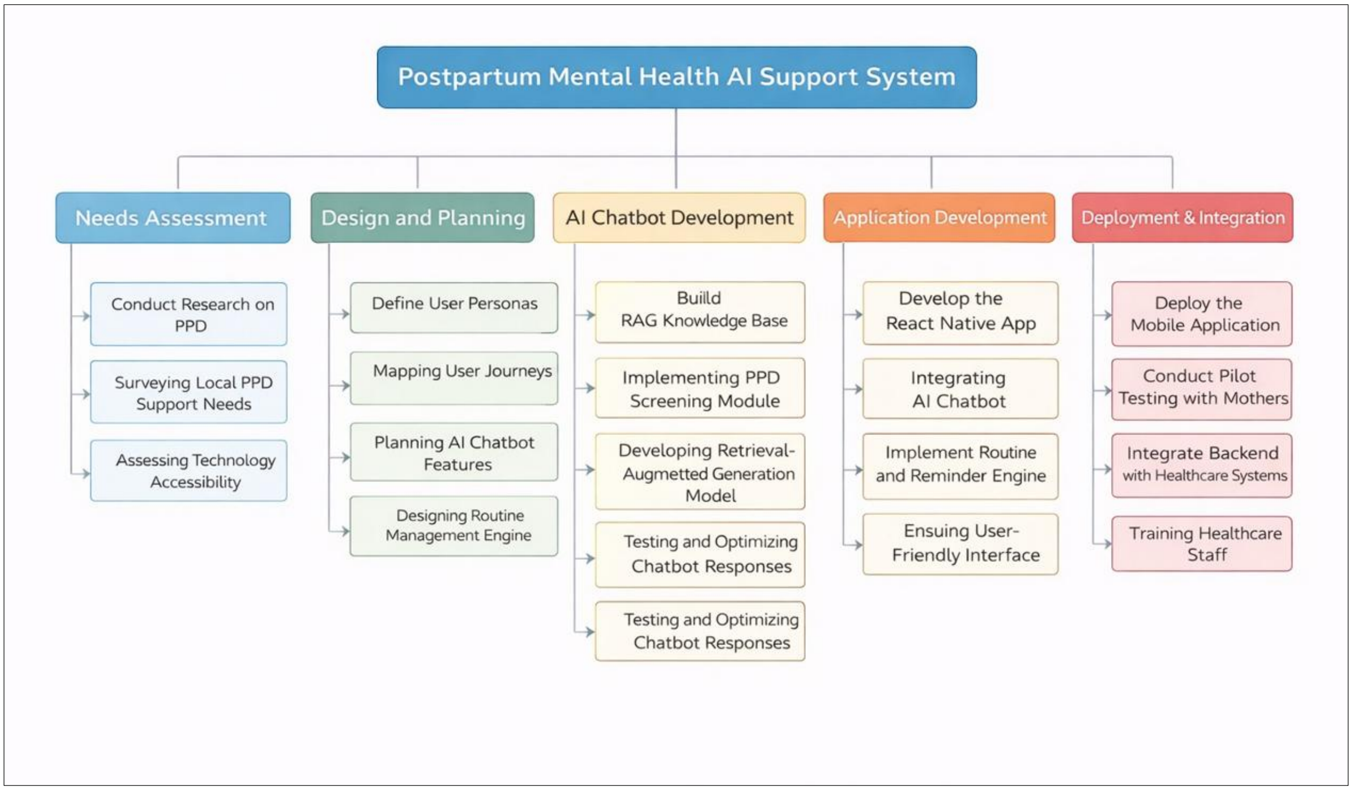

Figure 4 – Work Breakdown Structure of the Postpartum Mental Health AI Support System

Figure 4 shows the work breakdown of the proposed Postpartum Mental Health AI Support System. The development process is divided into five main stages: needs assessment, design and planning, AI chatbot development, application development, and deployment with integration.

First, the needs assessment stage focuses on researching postpartum depression, identifying local support needs, and understanding technology accessibility among mothers.

Next, the design and planning stage defines user personas, maps user journeys, and plans the chatbot features and routine management functions.

The AI chatbot development stage involves building the RAG-based knowledge base, implementing the PPD screening module, and developing the chatbot response system.

The application development stage focuses on developing the React Native mobile application, integrating the chatbot, and implementing the routine and reminder features.

Finally, the deployment and integration stage includes deploying the mobile application, conducting pilot testing with mothers, and integrating the system with healthcare environments where possible.

9. GANTT CHART

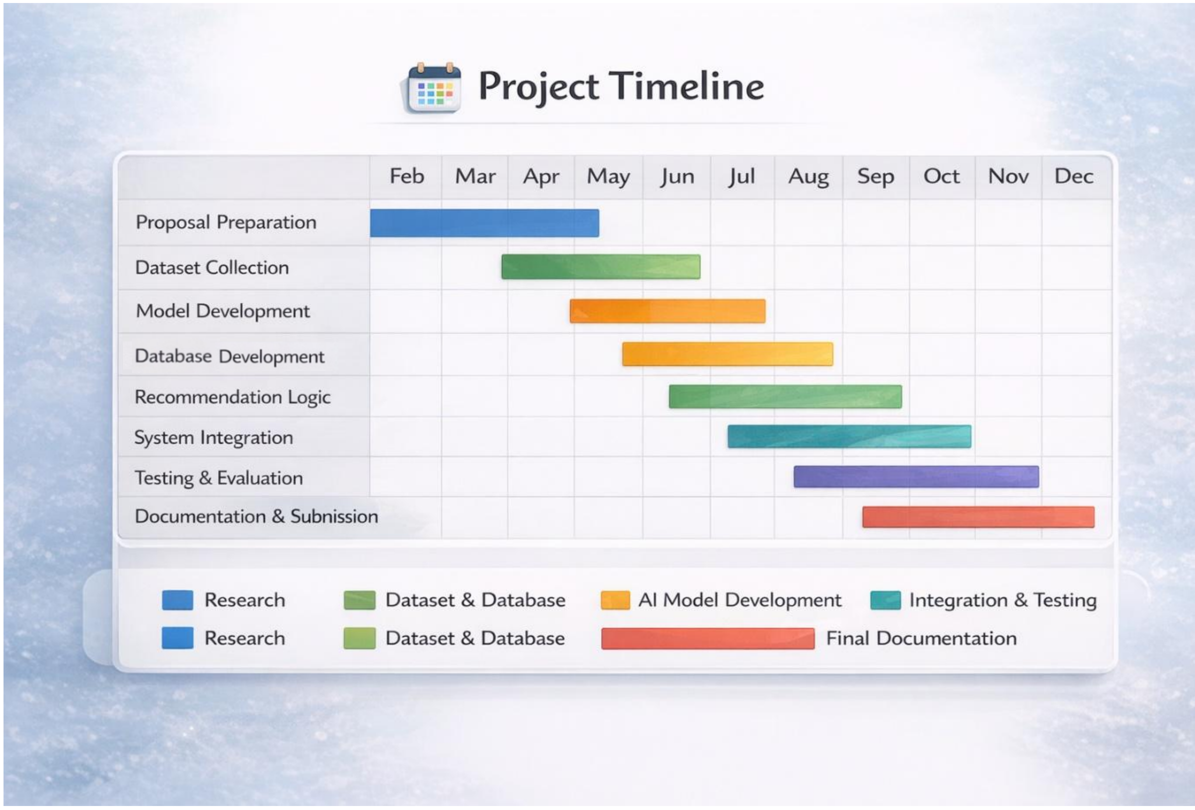

Figure 5 – Gantt Chart

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11. APPENDICES

Appendix - A: Work breakdown chart
Appendix - B: Gantt chart